Position-Time Graphs: Curved Motion (A-F)

Grade 10 / MYP 5 | Worked solutions

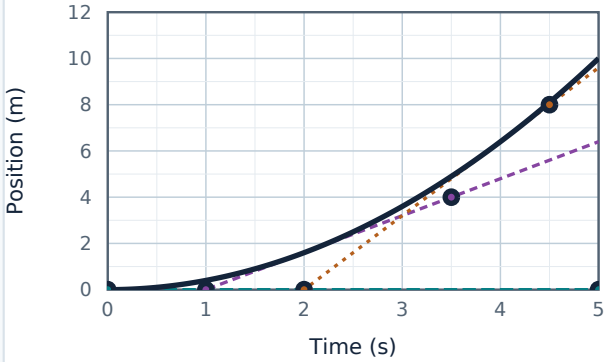
Gradient of tangent = instantaneous velocity. Acceleration = change in velocity / time.

All graphs have constant acceleration; the largest speed attained in each graph is 4 m/s.

Tangent key: t = 0 s t = 2 s t = 4 s Dots = exact grid points on tangents

Graph A

$x = 0.4t^2$



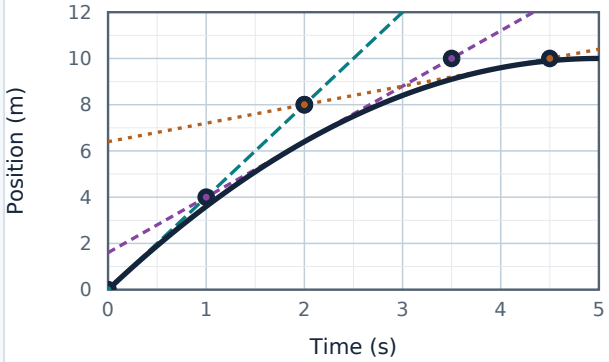
v at 0, 2, 4 s: 0, +1.6, +3.2 m/s

a = +0.8 m/s² (constant)

Moves forward, speeding up.

Graph B

$x = 4t - 0.4t^2$



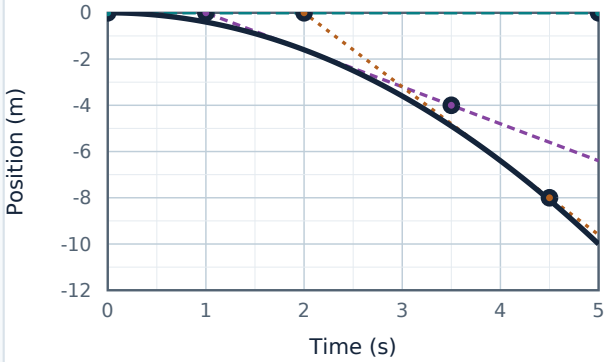
v at 0, 2, 4 s: +4.0, +2.4, +0.8 m/s

a = -0.8 m/s² (constant)

Moves forward, slowing to rest at 5 s.

Graph C

$x = -0.4t^2$



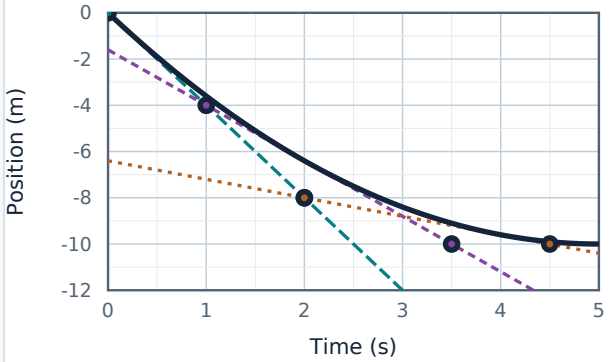
v at 0, 2, 4 s: 0, -1.6, -3.2 m/s

a = -0.8 m/s² (constant)

Moves backward, speeding up.

Graph D

$x = -4t + 0.4t^2$



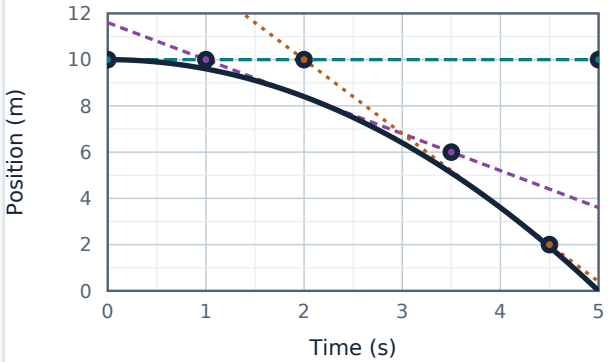
v at 0, 2, 4 s: -4.0, -2.4, -0.8 m/s

a = +0.8 m/s² (constant)

Moves backward, slowing to rest at 5 s.

Graph E

$x = 10 - 0.4t^2$



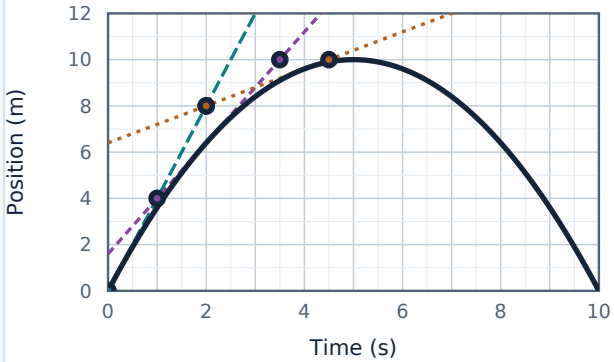
v at 0, 2, 4 s: 0, -1.6, -3.2 m/s

a = -0.8 m/s² (constant)

Moves backward from +10 m to 0 m, speeding up.

Graph F

$x = 4t - 0.4t^2$



v at 0, 2, 4 s: +4.0, +2.4, +0.8 m/s

a = -0.8 m/s² (constant)

Stops at 5 s; reverses direction; returns to 0 m.